Task:

1. Set up and configure an Nginx server on an EC2 instance using Amazon Linux, Ubuntu, and Red Hat.
2. Configure the instance with Nginx using user data (a Bash script). In both cases, a default web page must be displayed.

Expected Outcome:

- Nginx should be successfully running on all instances.
- Accessing the public IP in a browser should display the default web page.
- Automation using user data should eliminate the need for manual setup.

Set up and configure an Nginx server on an EC2 instance using Amazon Linux, Ubuntu, and Red Hat.

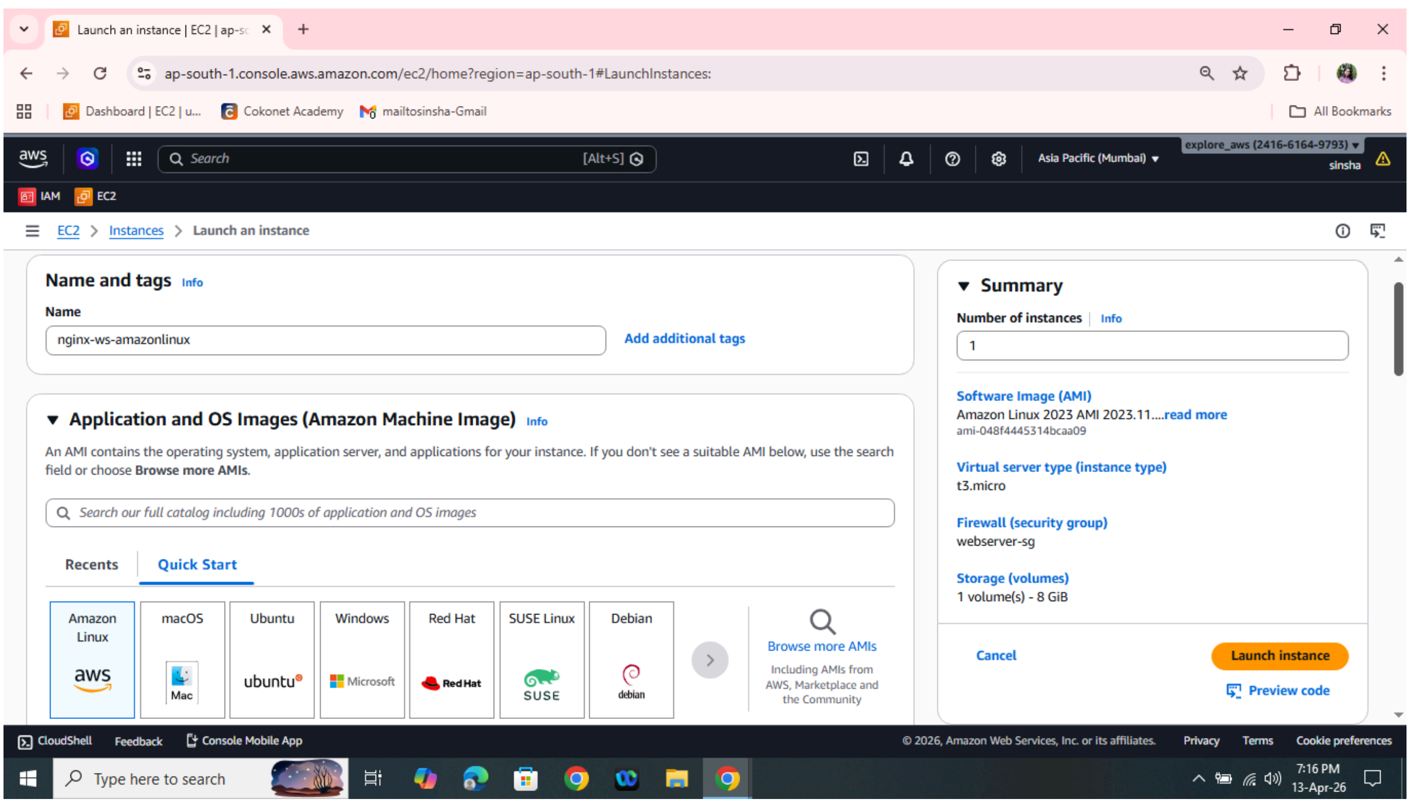
PREREQUISITES:

1. Login to AWS account in the preferred region (Eg: Mumbai)
2. Create a Key pair in selected region
3. Select EC2 service from the list of services
4. Create a security group by allowing SSH(Port: 22) and HTTP(Port: 80) from Network & Security section on the left panel of EC2.
5. Then back to EC2 and click Launch Instance.
6. Then create EC2 instances with mentioned AMIs.
    a. Amazon Linux 2
    b. Ubuntu
    c. Red Hat Enterprise Linux

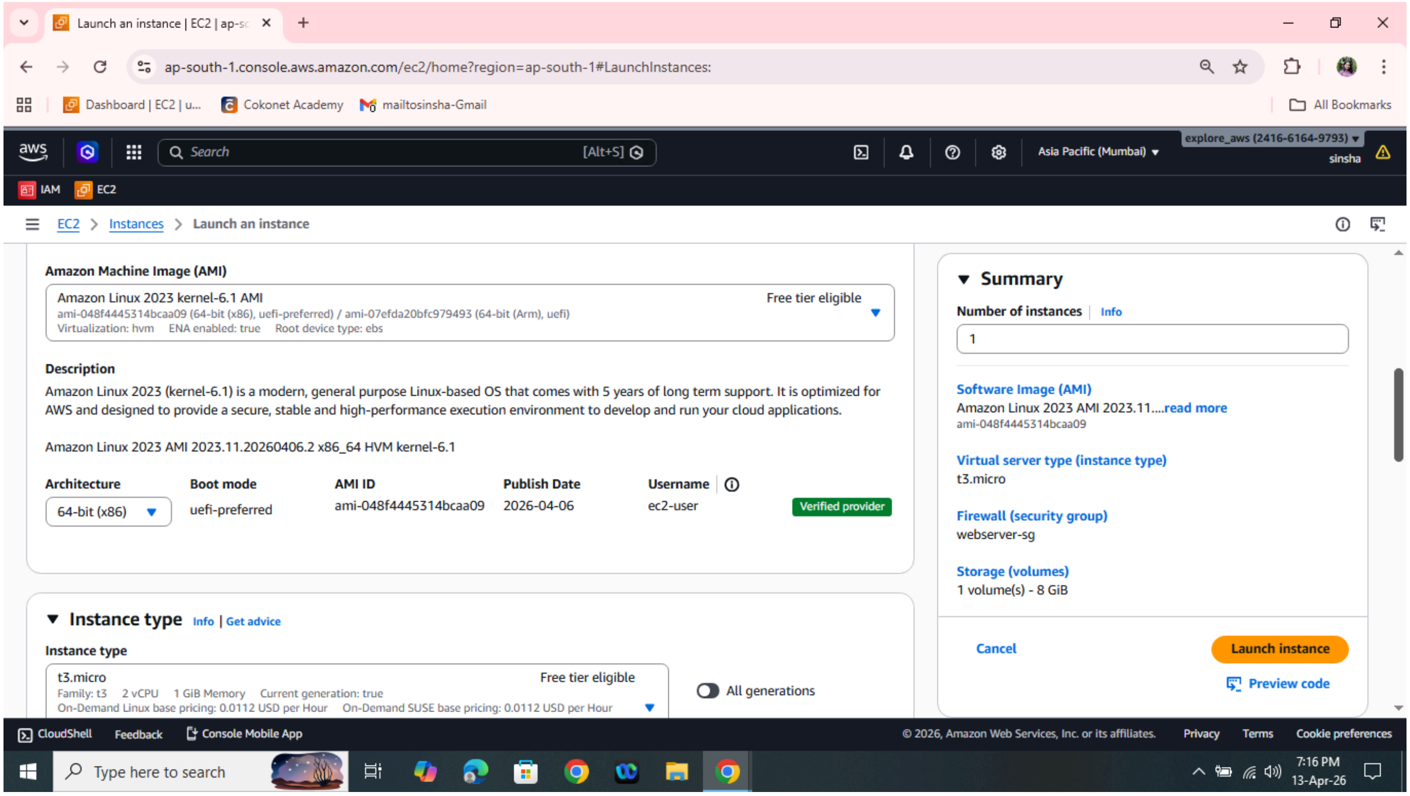
CONFIGURATION STEPS:

1. AMAZON LINUX

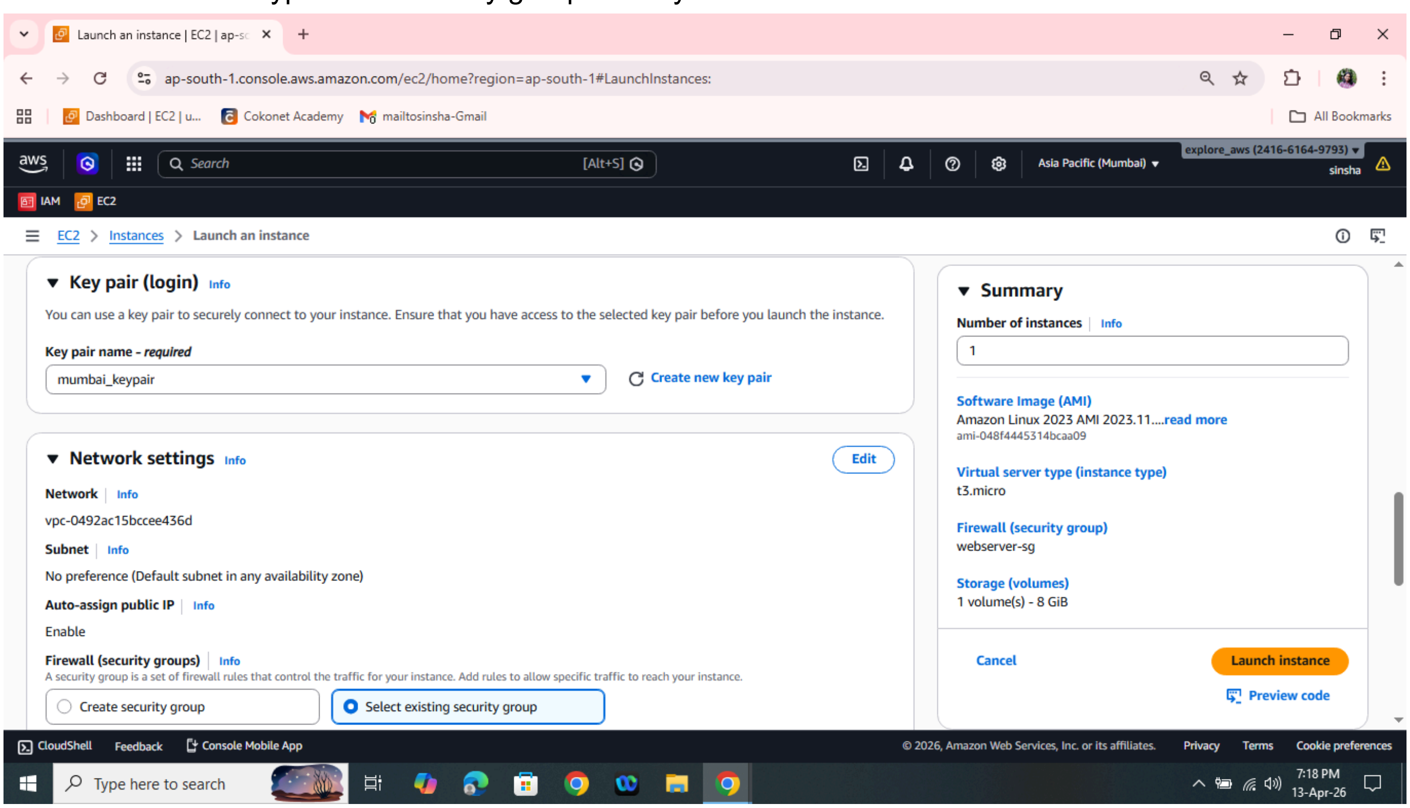
- Name the instance and select OS Image



- Select AMI (free tier eligible for test purpose)

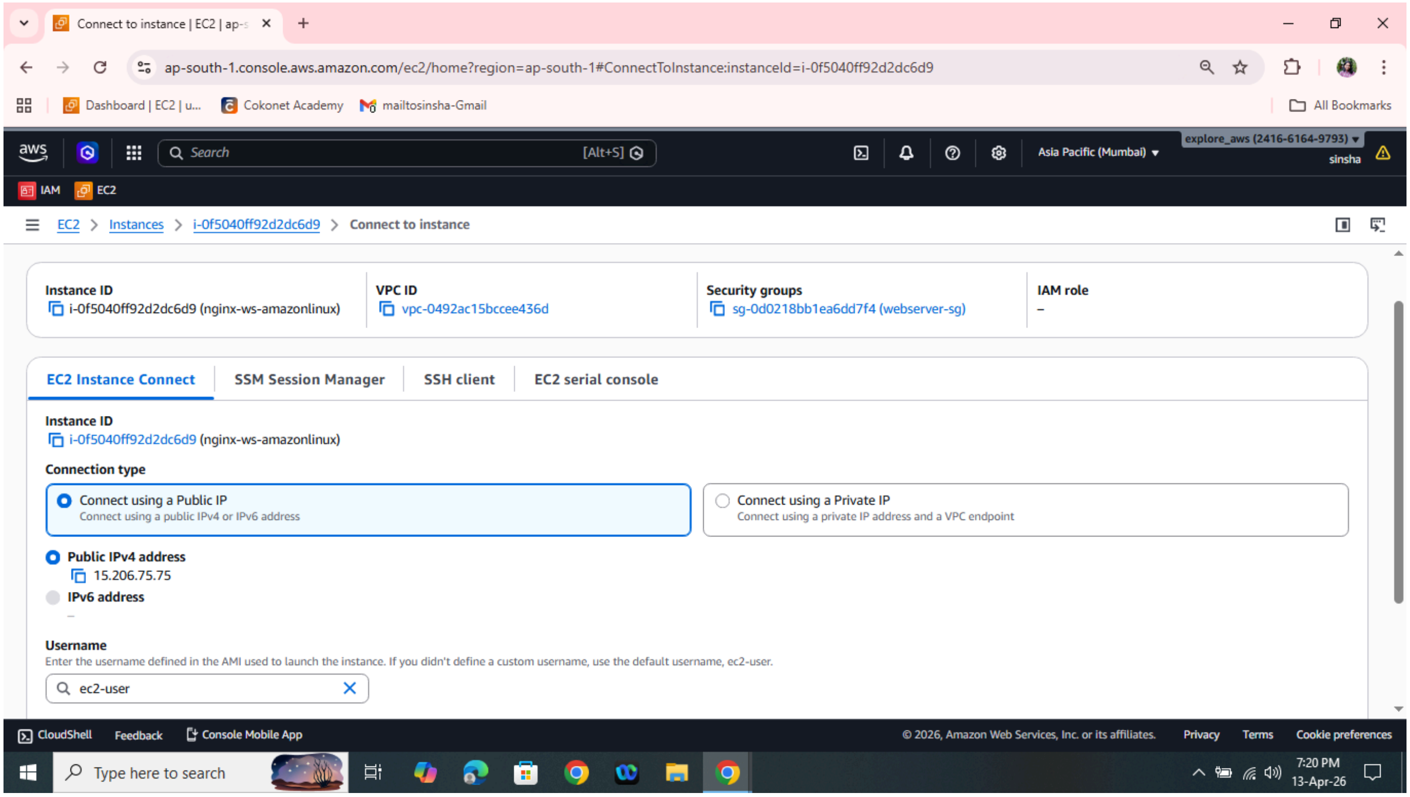


- Select the Keypair and security group already created



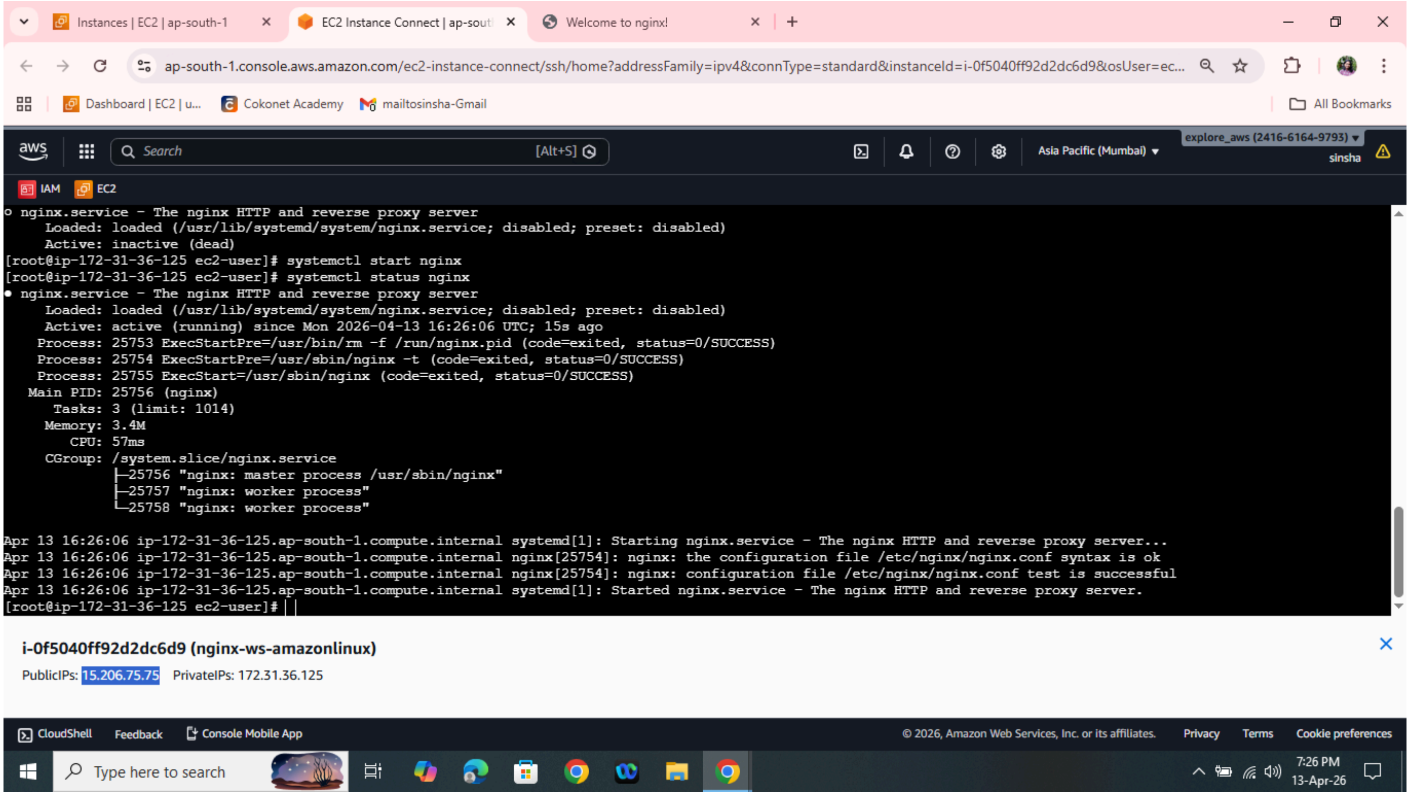
- Then click on Launch instance.

- After an instance has been created login into Amazon Linux instance by Amazon browser based login as shown below.
- In the case of Redhat use Mobaxterm or Putty for login to the terminal.

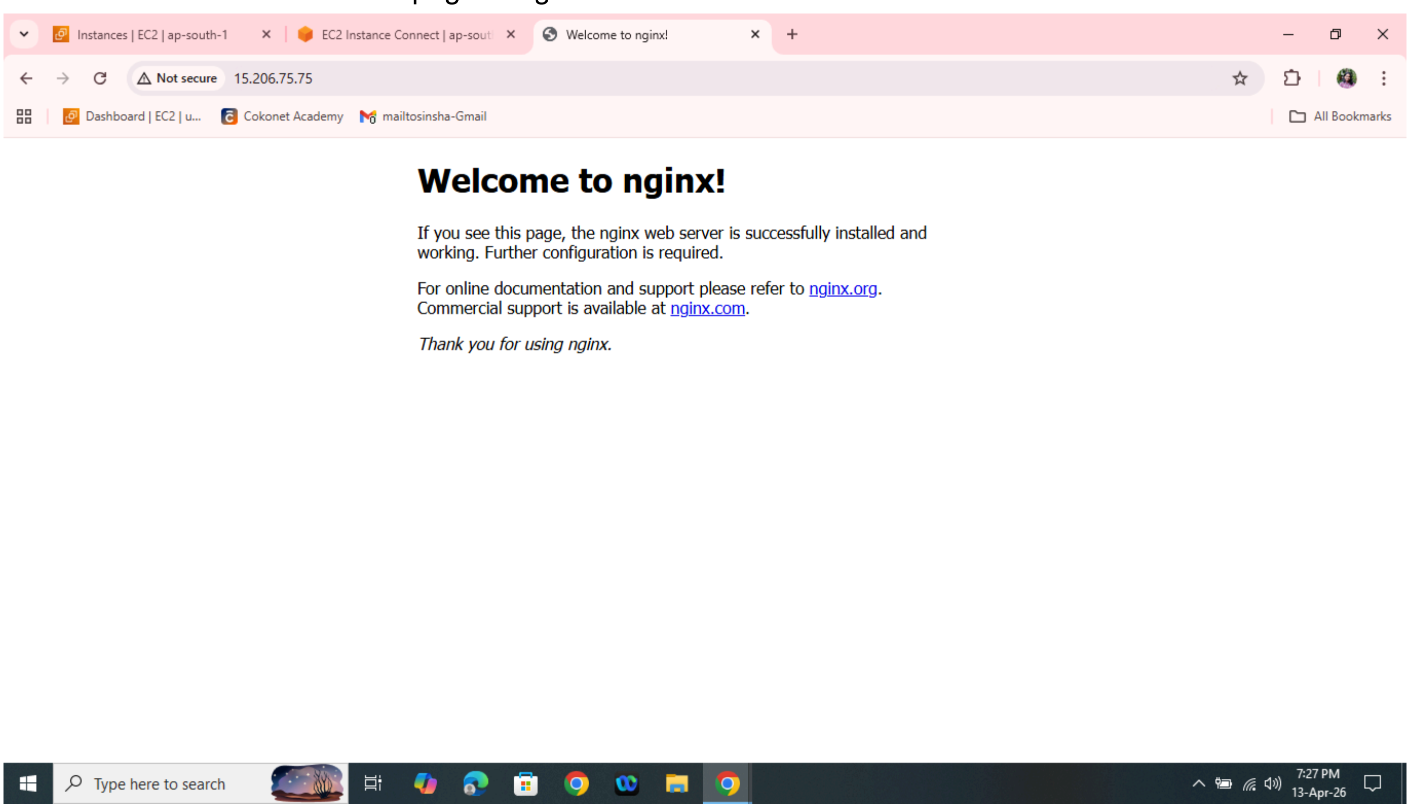


Do the following command execution on the coming terminal

1. sudo su
2. dnf update -y
3. dnf install nginx -y
4. systemctl start nginx

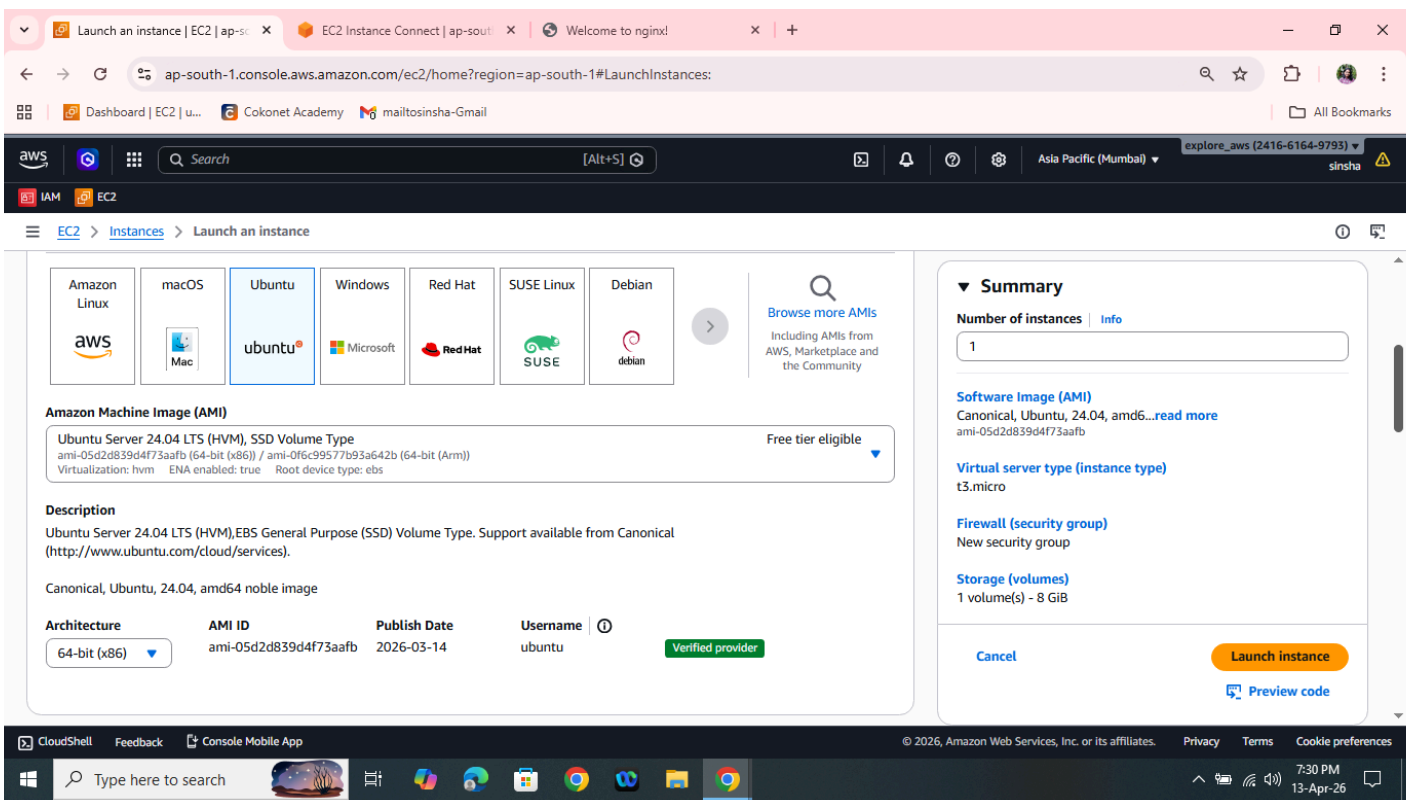


- After nginx service is successfully running, copy the public IP of the instance and type in any browser as below:
    - http://<IP ADDRESS OF THE INSTANCE>/
    - Then the default page of nginx will be shown below.



- Do the same step for other mentioned os as well
- The execution command will be different for Ubuntu.

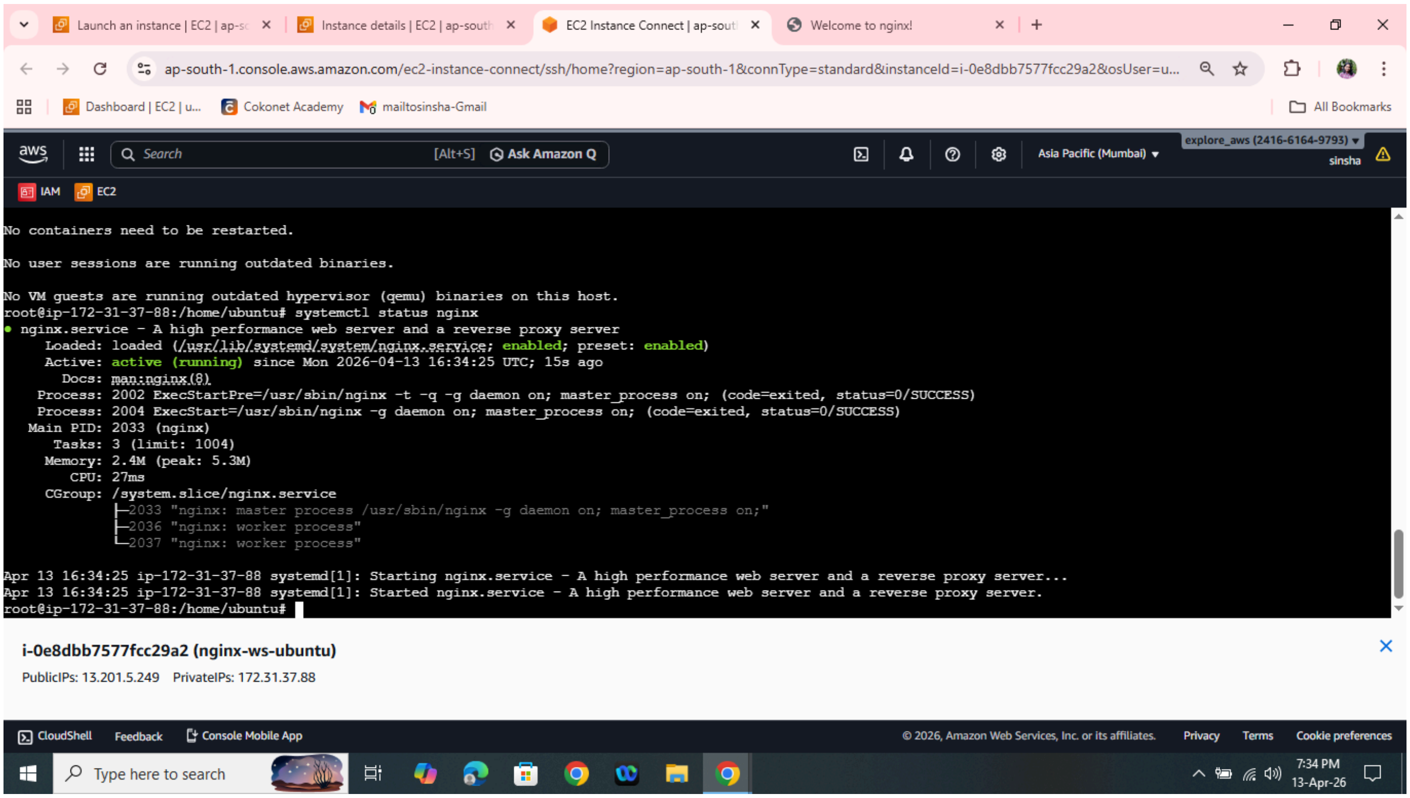
2. UBUNTU

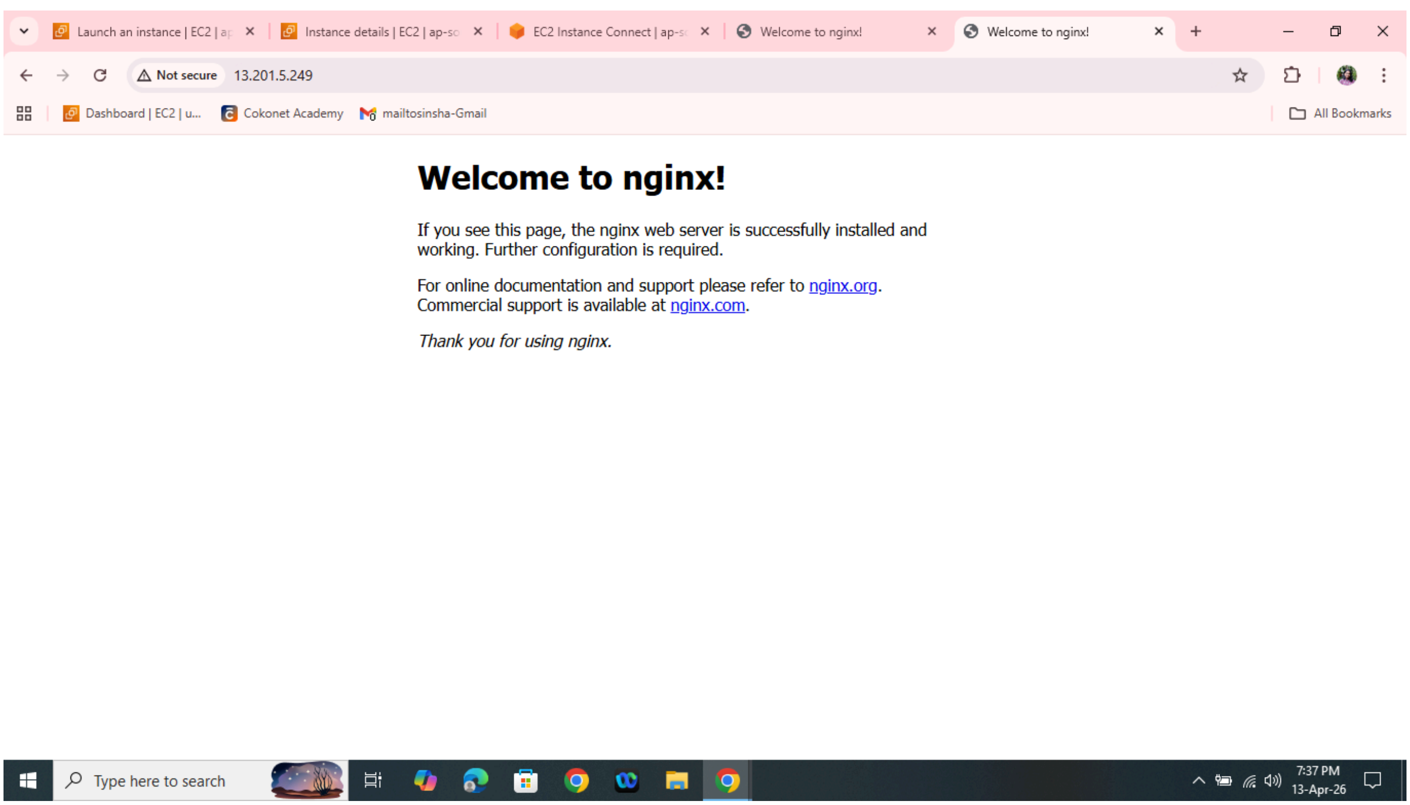


Commands executed:

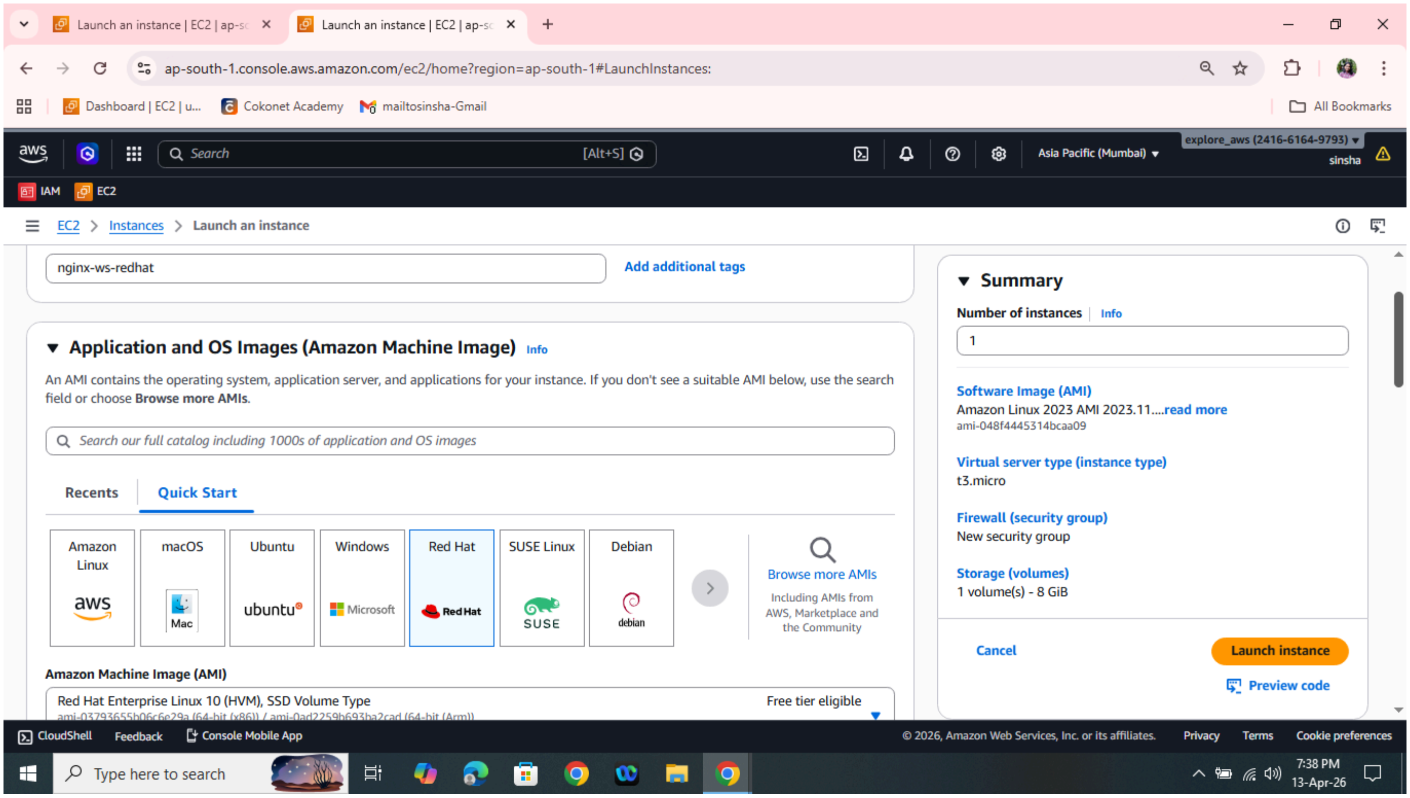
1. sudo su
2. apt update
3. apt install nginx -y

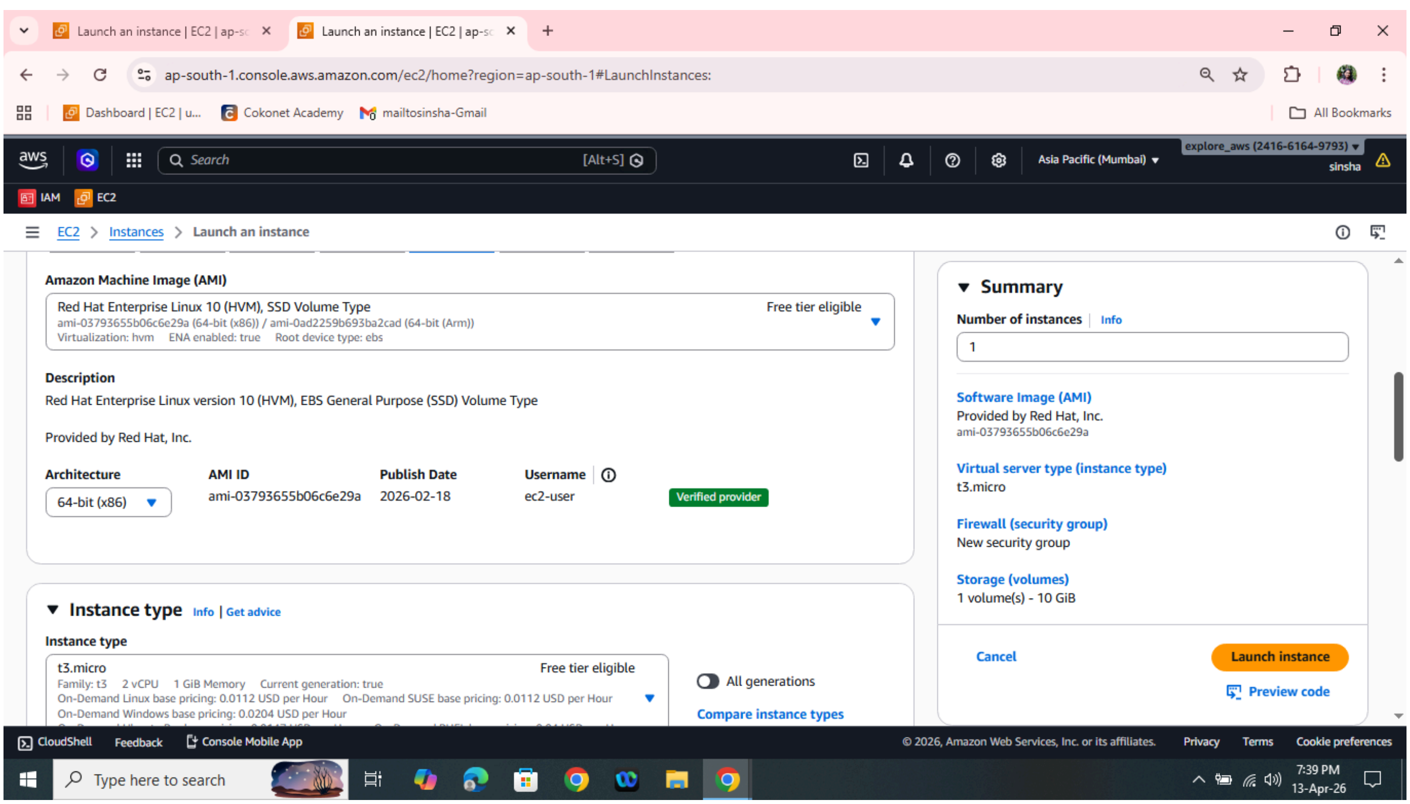
4. systemctl status nginx
5. Nginx will be running state by default
6. Start nginx cmd: systemctl start nginx

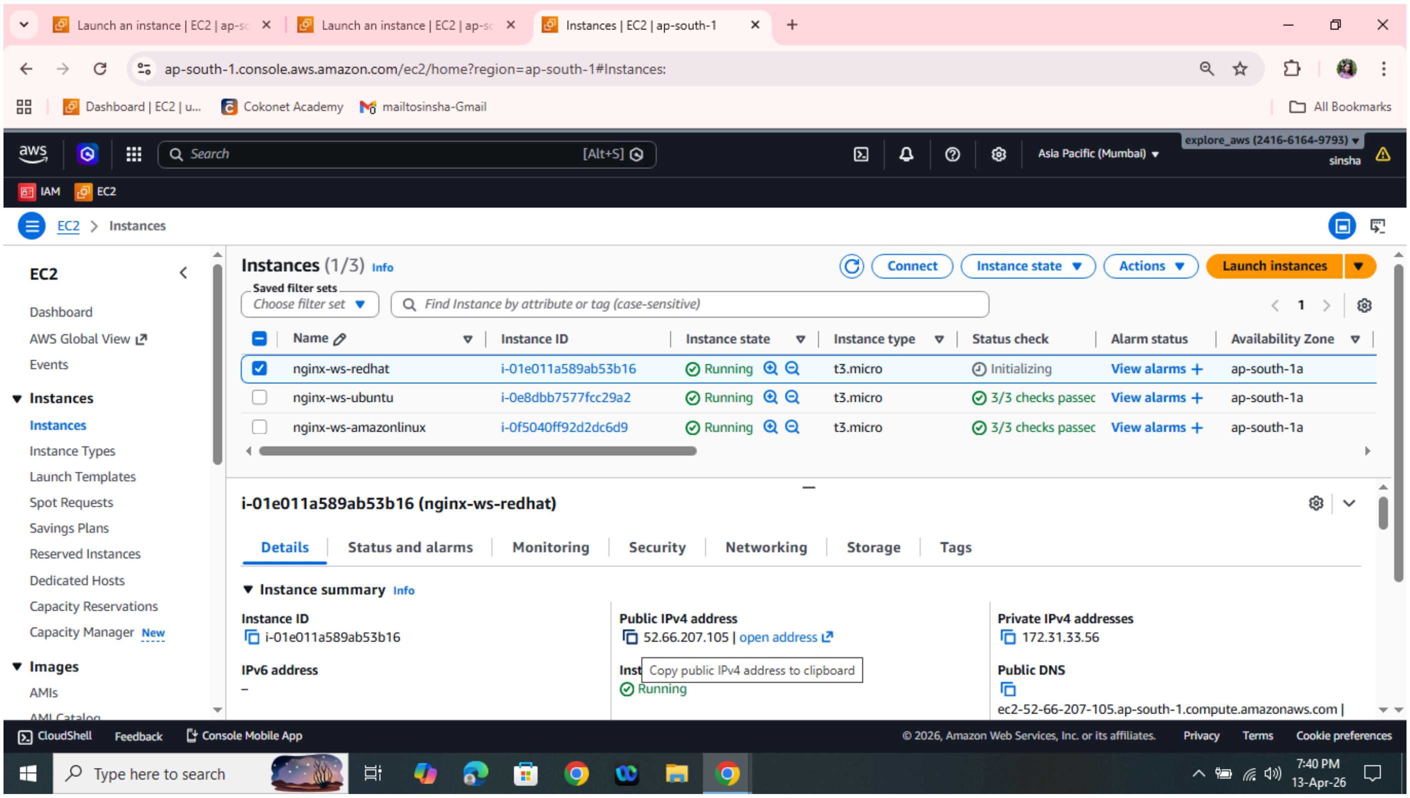


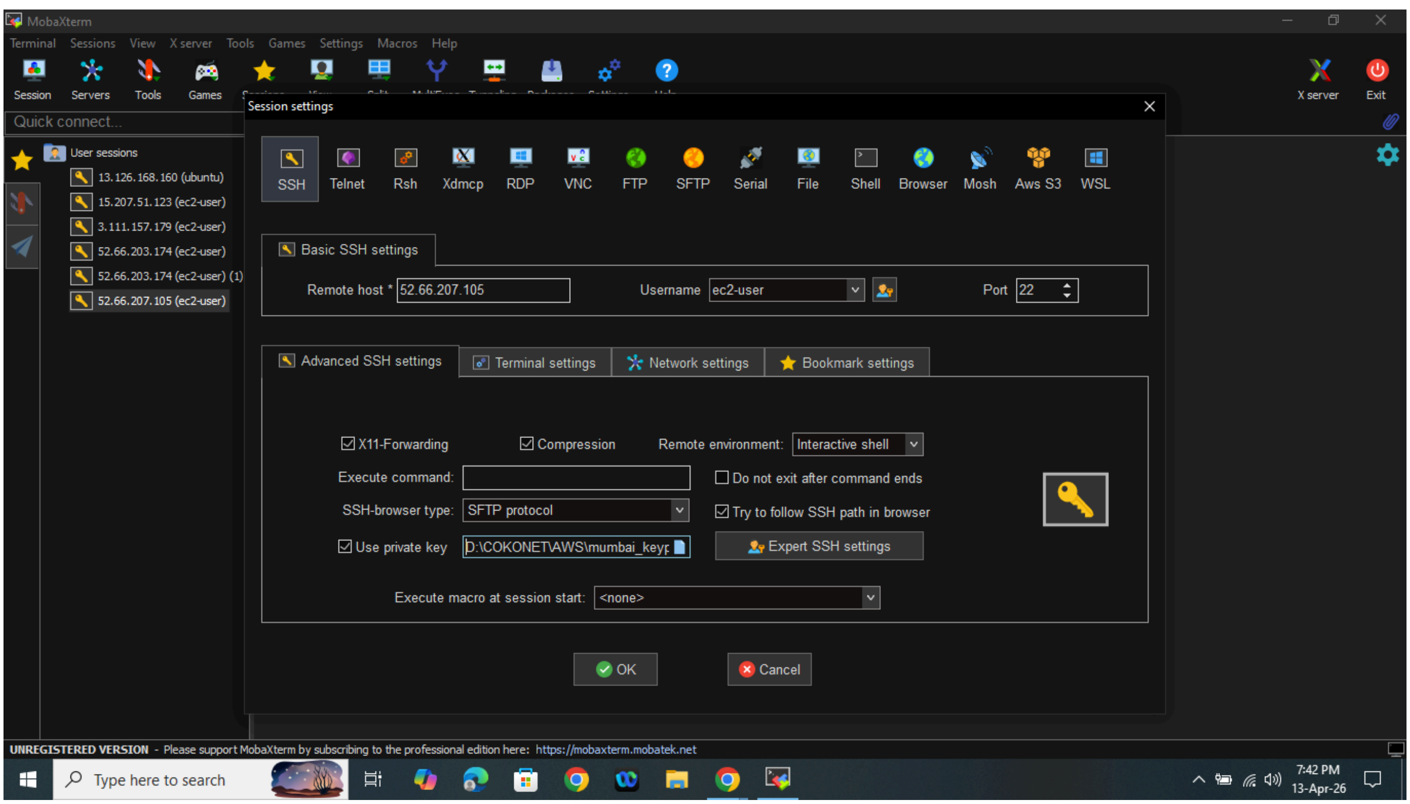


3. REDHAT


Launch an instance | EC2 | ap-s
Launch an instance | EC2 | ap-s
ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#LaunchInstances:
Dashboard | EC2 | u...
Cokonet Academy
mailtosinsha-Gmail
All Bookmarks
Search
[Alt+S]
Asia Pacific (Mumbai)
explore_aws (2416-6164-9793)
sinsha
IAM
EC2
EC2 > Instances > Launch an instance
nginx-ws-redhat
Add additional tags
Application and OS Images (Amazon Machine Image) Info
An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose Browse more AMIs.
Search our full catalog including 1000s of application and OS images
Recents
Quick Start
Amazon Linux
macOS
Ubuntu
Windows
Red Hat
SUSE Linux
Debian
Browse more AMIs
Including AMIs from AWS, Marketplace and the Community
Amazon Machine Image (AMI)
Red Hat Enterprise Linux 10 (HVM), SSD Volume Type
Free tier eligible
Summary
Number of instances Info
1
Software Image (AMI)
Amazon Linux 2023 AMI 2023.11....read more
ami-048f4445314bcaa09
Virtual server type (instance type)
t3.micro
Firewall (security group)
New security group
Storage (volumes)
1 volume(s) - 8 GiB
Cancel
Launch instance
Preview code
CloudShell
Feedback
Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences
Type here to search
7:38 PM
13-Apr-26

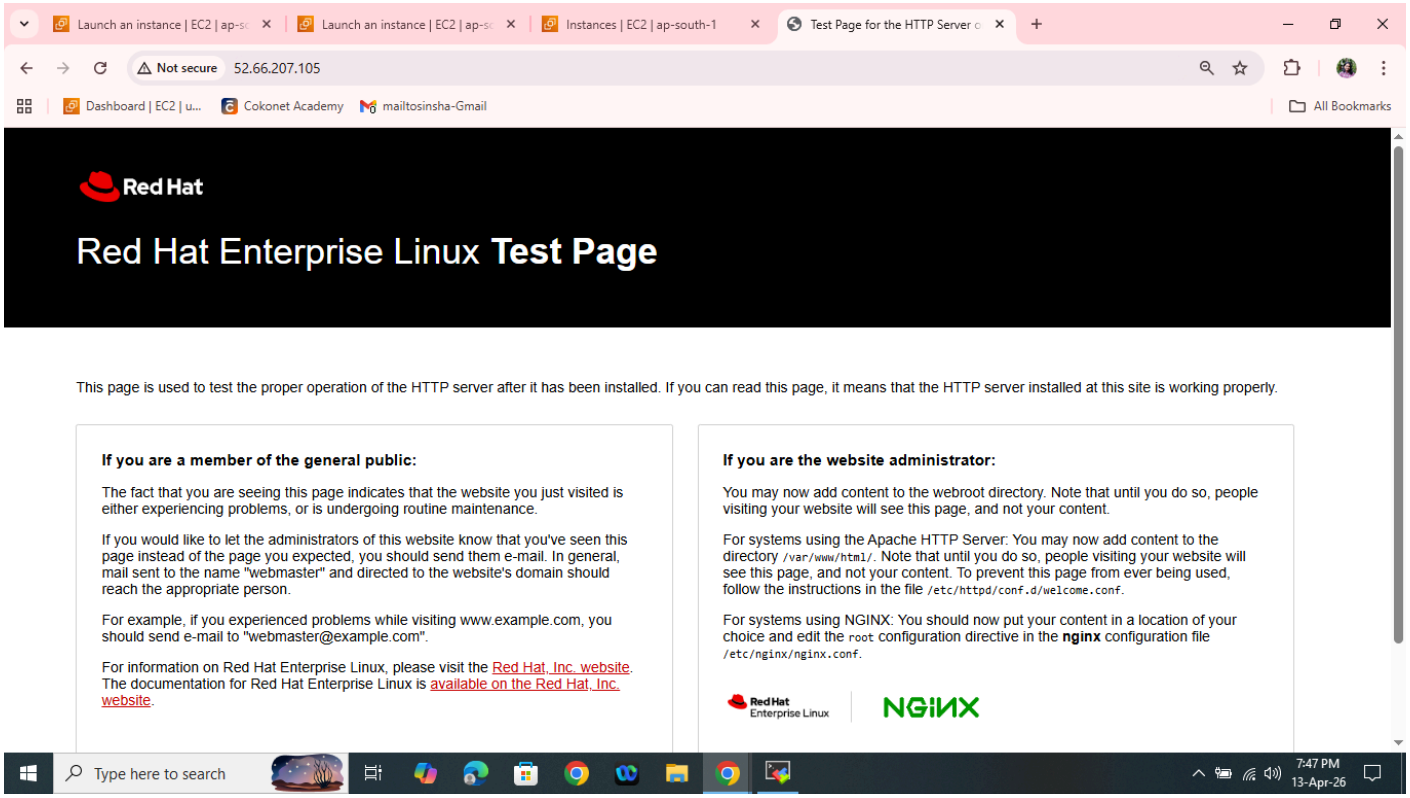

Launch an instance | EC2 | ap-s
Launch an instance | EC2 | ap-s
ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#LaunchInstances:
Dashboard | EC2 | u...
Cokonet Academy
mailtosinsha-Gmail
All Bookmarks
Search
[Alt+S]
Asia Pacific (Mumbai)
explore_aws (2416-6164-9793)
sinsha
IAM
EC2
EC2 > Instances > Launch an instance
Amazon Machine Image (AMI)
Red Hat Enterprise Linux 10 (HVM), SSD Volume Type
Free tier eligible
ami-03793655b06c6e29a (64-bit (x86)) / ami-0ad2259b693ba2cad (64-bit (Arm))
Virtualization: hvm ENA enabled: true Root device type: ebs
Description
Red Hat Enterprise Linux version 10 (HVM), EBS General Purpose (SSD) Volume Type
Provided by Red Hat, Inc.
Architecture
64-bit (x86)
AMI ID
ami-03793655b06c6e29a
Publish Date
2026-02-18
Username
ec2-user
Verified provider
Instance type Info | Get advice
Instance type
t3.micro
Free tier eligible
Family: t3 2 vCPU 1 GiB Memory Current generation: true
On-Demand Linux base pricing: 0.0112 USD per Hour On-Demand SUSE base pricing: 0.0112 USD per Hour
On-Demand Windows base pricing: 0.0204 USD per Hour
All generations
Compare instance types
Summary
Number of instances Info
1
Software Image (AMI)
Provided by Red Hat, Inc.
ami-03793655b06c6e29a
Virtual server type (instance type)
t3.micro
Firewall (security group)
New security group
Storage (volumes)
1 volume(s) - 10 GiB
Cancel
Launch instance
Preview code
CloudShell
Feedback
Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences
Type here to search
7:39 PM
13-Apr-26


Launch an instance | EC2 | ap-so
Launch an instance | EC2 | ap-so
Instances | EC2 | ap-south-1
ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#Instances:
Dashboard | EC2 | u...
Cokonet Academy
mailtosinsha-Gmail
All Bookmarks
Search
[Alt+S]
Asia Pacific (Mumbai)
explore_aws (2416-6164-9793)
sinsha
IAM
EC2
EC2 > Instances
EC2
Dashboard
AWS Global View
Events
Instances
Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
Capacity Manager New
Images
AMIs
Instances (1/3) Info
Connect
Instance state
Actions
Launch instances
Saved filter sets
Choose filter set
Find Instance by attribute or tag (case-sensitive)
Name
Instance ID
Instance state
Instance type
Status check
Alarm status
Availability Zone
nginx-ws-redhat i-01e011a589ab53b16 Running t3.micro Initializing View alarms ap-south-1a
nginx-ws-ubuntu i-0e8dbb7577fcc29a2 Running t3.micro 3/3 checks passec View alarms ap-south-1a
nginx-ws-amazonlinux i-0f5040ff92d2dc6d9 Running t3.micro 3/3 checks passec View alarms ap-south-1a
i-01e011a589ab53b16 (nginx-ws-redhat)
Details
Status and alarms
Monitoring
Security
Networking
Storage
Tags
Instance summary Info
Instance ID
i-01e011a589ab53b16
Public IPv4 address
52.66.207.105 | open address
Private IPv4 addresses
172.31.33.56
IPv6 address
Copy public IPv4 address to clipboard
Running
Public DNS
ec2-52-66-207-105.ap-south-1.compute.amazonaws.com
CloudShell
Feedback
Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences
Type here to search
7:40 PM
13-Apr-26


MobaXterm
Terminal Sessions View X server Tools Games Settings Macros Help
Session
Servers
Tools
Games
X server
Exit
Quick connect...
User sessions
13.126.168.160 (ubuntu)
15.207.51.123 (ec2-user)
3.111.157.179 (ec2-user)
52.66.203.174 (ec2-user)
52.66.203.174 (ec2-user) (1)
52.66.207.105 (ec2-user)
Session settings
SSH
Telnet
Rsh
Xdmcp
RDP
VNC
FTP
SFTP
Serial
File
Shell
Browser
Mosh
Aws S3
WSL
Basic SSH settings
Remote host * 52.66.207.105
Username ec2-user
Port 22
Advanced SSH settings
Terminal settings
Network settings
Bookmark settings
X11-Forwarding
Compression
Remote environment: Interactive shell
Execute command:
Do not exit after command ends
SSH-browser type: SFTP protocol
Try to follow SSH path in browser
Use private key D:\COKONET\AWS\mumbai_keyp
Expert SSH settings
Execute macro at session start: <none>
OK
Cancel
UNREGISTERED VERSION - Please support MobaXterm by subscribing to the professional edition here: https://mobaxterm.mobatek.net
Type here to search
7:42 PM
13-Apr-26

Command executed:

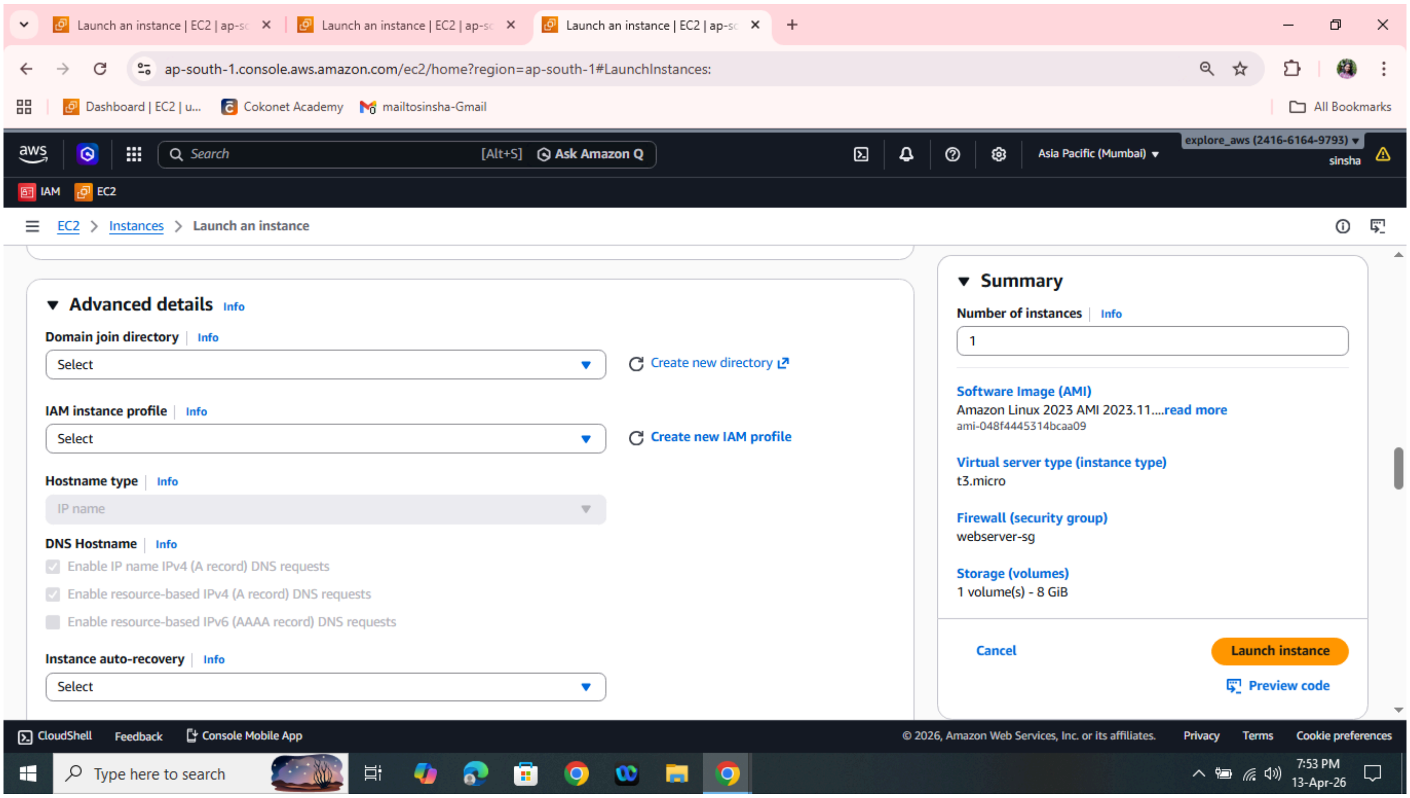
1. dnf update
2. dnf install nginx -y
3. systemctl status nginx
4. systemctl start nginx
5. systemctl status nginx

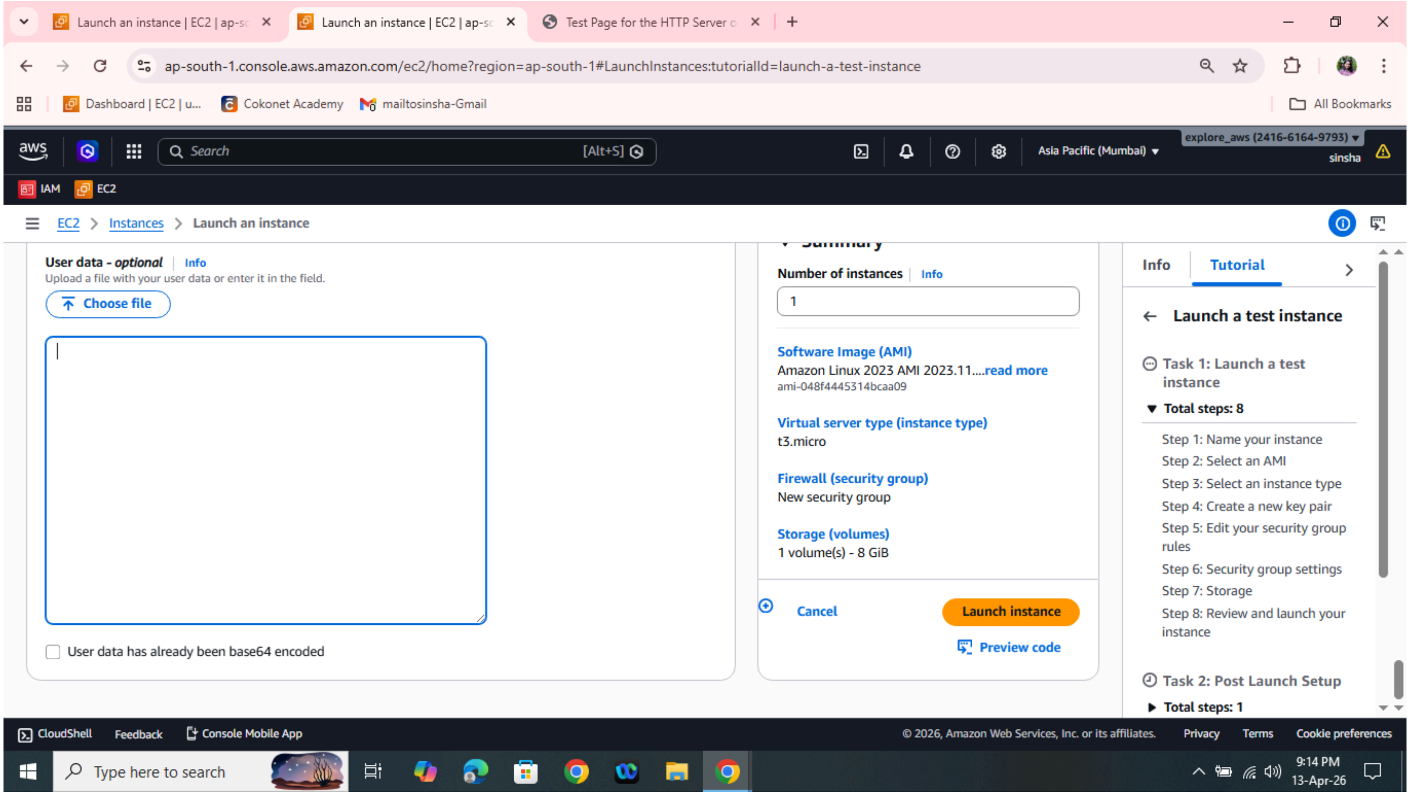


Configure the instance with Nginx using user data (a Bash script). In both cases, a default web page must be displayed.

CONFIGURATION STEPS:

1. Prerequisites will be the same as above.
2. Click on Launch instance from EC2 service from aws
3. Name the instance and select OS Image, instance type, Key pair and Security group
4. Then click on Advance details and come down where you see ***User data***

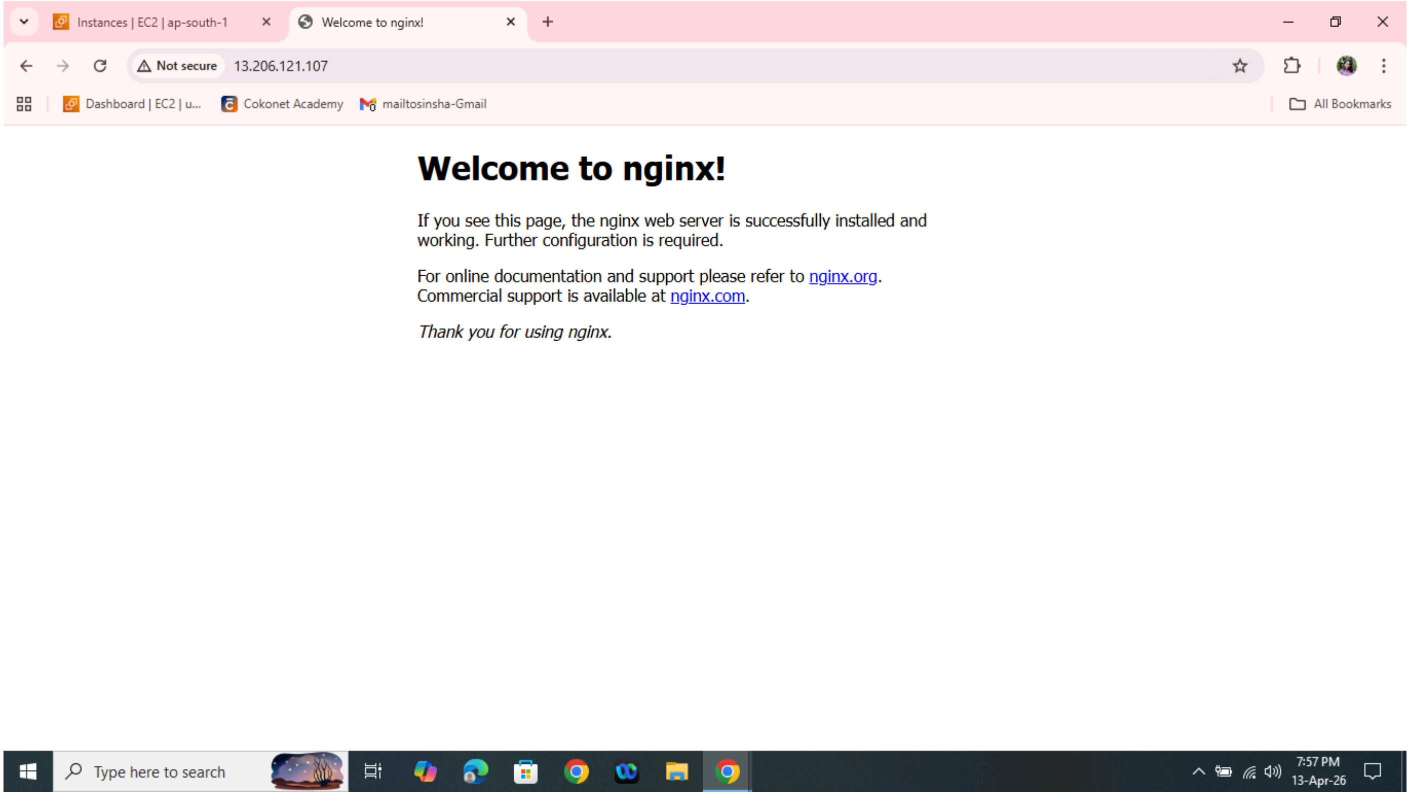




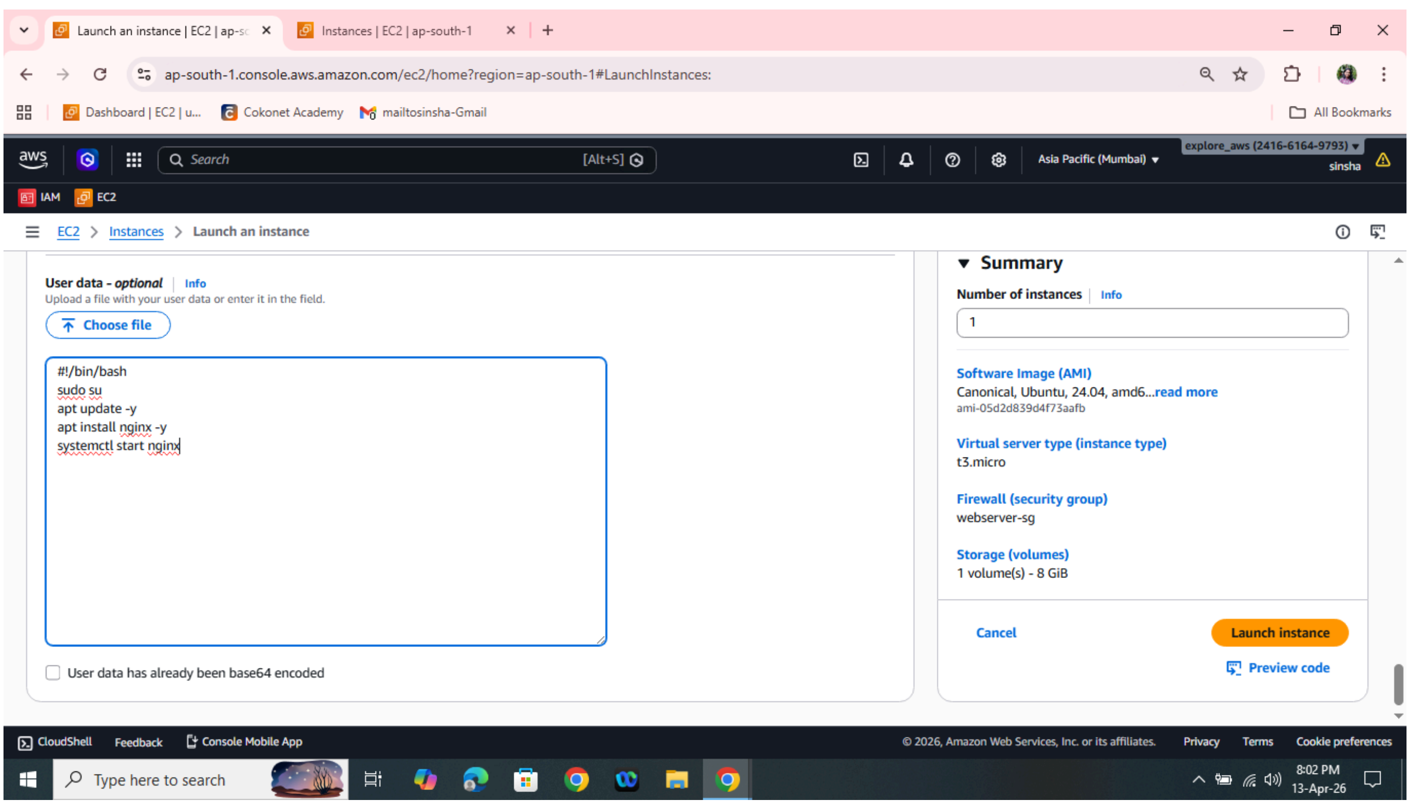
5. Fill the user data details according to the AMI you have selected, then Launch instance.
6. Once the instance is launched open the public ip the browser, where you can see the default page nginx

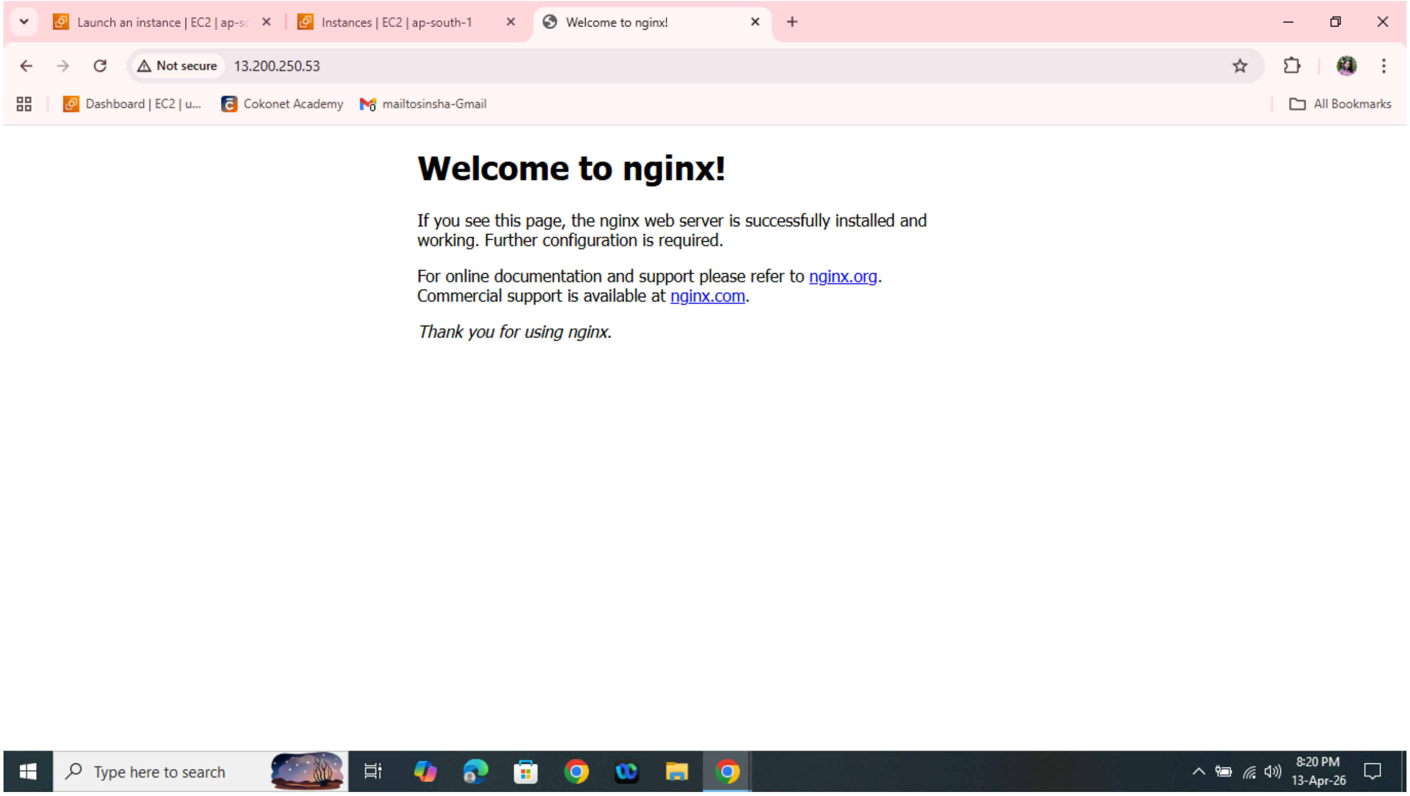
For AMAZON LINUX

```
#!/bin/bash
sudo su
dnf install nginx -y
systemctl start nginx
```

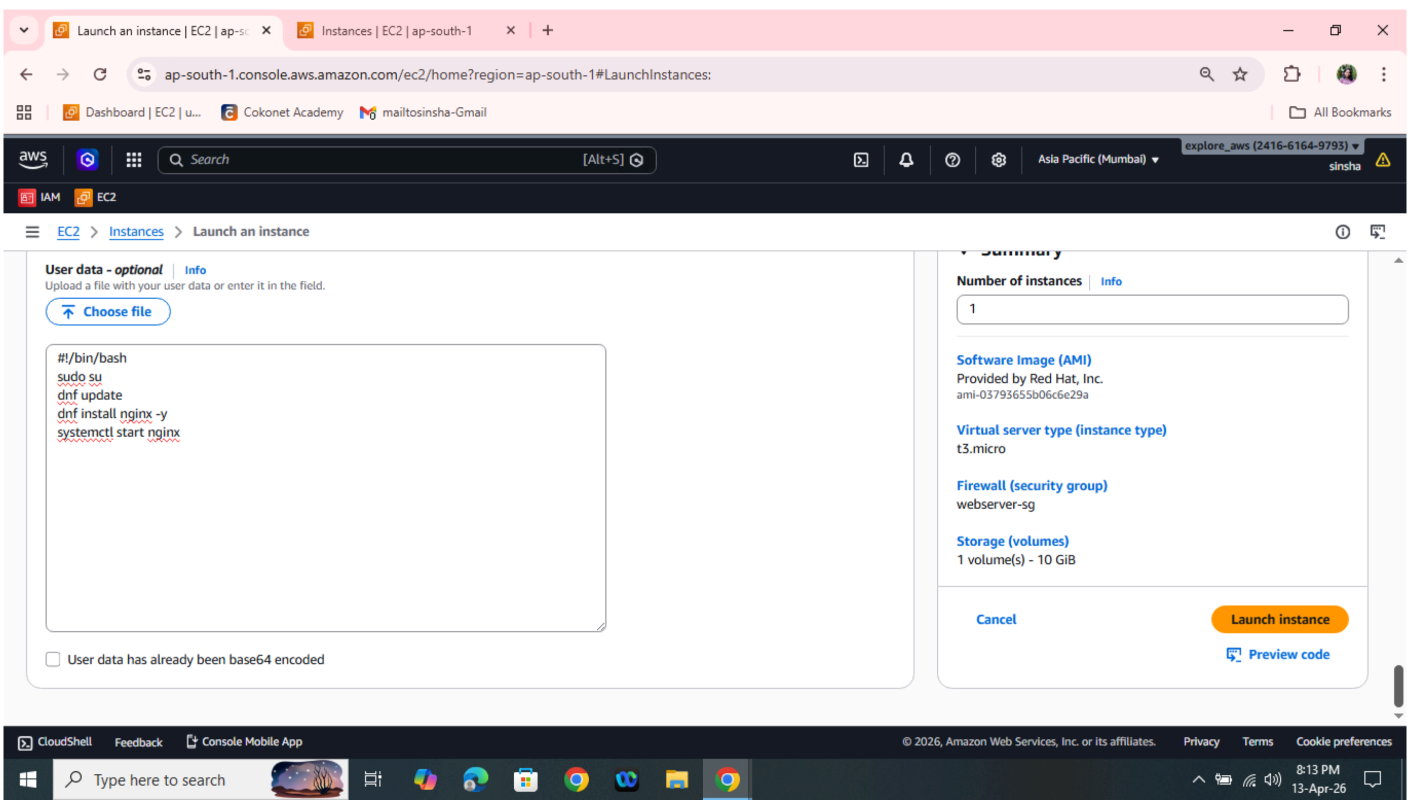

Instances | EC2 | ap-south-1
Welcome to nginx!
Not secure
13.206.121.107
Dashboard | EC2 | u...
Cokonet Academy
mailtosinsha-Gmail
All Bookmarks
Welcome to nginx!
If you see this page, the nginx web server is successfully installed and working. Further configuration is required.
For online documentation and support please refer to nginx.org.
Commercial support is available at nginx.com.
Thank you for using nginx.
Type here to search
7:57 PM
13-Apr-26

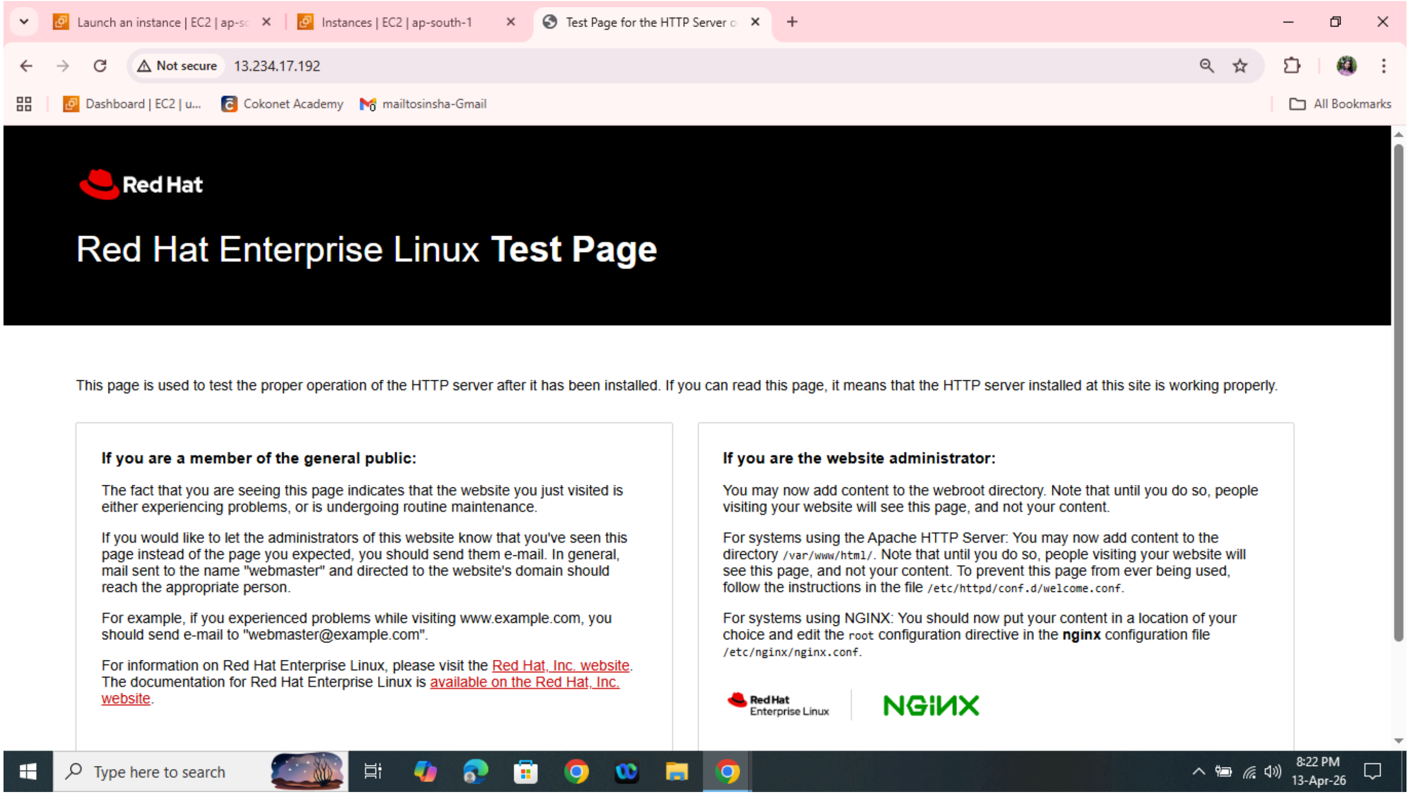
2. UBUNTU


Launch an instance | EC2 | ap-s
Instances | EC2 | ap-south-1
ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#LaunchInstances:
Dashboard | EC2 | u...
Cokonet Academy
mailtosinsha-Gmail
All Bookmarks
Search
[Alt+S]
Asia Pacific (Mumbai)
explore_aws (2416-6164-9793)
sinsha
IAM
EC2
EC2 > Instances > Launch an instance
User data - optional
Info
Upload a file with your user data or enter it in the field.
Choose file
#!/bin/bash
sudo su
apt update -y
apt install nginx -y
systemctl start nginx
User data has already been base64 encoded
Summary
Number of instances
Info
1
Software Image (AMI)
Canonical, Ubuntu, 24.04, amd6...read more
ami-05d2d839d4f73aafb
Virtual server type (instance type)
t3.micro
Firewall (security group)
webserver-sg
Storage (volumes)
1 volume(s) - 8 GiB
Cancel
Launch instance
Preview code
CloudShell
Feedback
Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences
Type here to search
8:02 PM
13-Apr-26


Launch an instance | EC2 | ap-s
Instances | EC2 | ap-south-1
Welcome to nginx!
Not secure
13.200.250.53
Dashboard | EC2 | u...
Cokonet Academy
mailtosinsha-Gmail
All Bookmarks
Welcome to nginx!
If you see this page, the nginx web server is successfully installed and working. Further configuration is required.
For online documentation and support please refer to nginx.org.
Commercial support is available at nginx.com.
Thank you for using nginx.
Type here to search
8:20 PM
13-Apr-26

3. REDHAT


Launch an instance | EC2 | ap-s
Instances | EC2 | ap-south-1
ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#LaunchInstances:
Dashboard | EC2 | u...
Cokonet Academy
mailtosinsha-Gmail
All Bookmarks
aws
Search
[Alt+S]
Asia Pacific (Mumbai)
explore_aws (2416-6164-9793)
sinsha
IAM
EC2
EC2
Instances
Launch an instance
User data - optional
Info
Upload a file with your user data or enter it in the field.
Choose file
#!/bin/bash
sudo su
dnf update
dnf install nginx -y
systemctl start nginx
User data has already been base64 encoded
Number of instances
Info
1
Software Image (AMI)
Provided by Red Hat, Inc.
ami-03793655b06c6e29a
Virtual server type (instance type)
t3.micro
Firewall (security group)
webserver-sg
Storage (volumes)
1 volume(s) - 10 GiB
Cancel
Launch instance
Preview code
CloudShell
Feedback
Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences
Type here to search
8:13 PM
13-Apr-26


Launch an instance | EC2 | ap-s
Instances | EC2 | ap-south-1
Test Page for the HTTP Server o
Not secure 13.234.17.192
Dashboard | EC2 | u...
Cokonet Academy
mailtosinsha-Gmail
All Bookmarks
Red Hat
Red Hat Enterprise Linux Test Page
This page is used to test the proper operation of the HTTP server after it has been installed. If you can read this page, it means that the HTTP server installed at this site is working properly.
If you are a member of the general public:
The fact that you are seeing this page indicates that the website you just visited is either experiencing problems, or is undergoing routine maintenance.
If you would like to let the administrators of this website know that you've seen this page instead of the page you expected, you should send them e-mail. In general, mail sent to the name "webmaster" and directed to the website's domain should reach the appropriate person.
For example, if you experienced problems while visiting www.example.com, you should send e-mail to "webmaster@example.com".
For information on Red Hat Enterprise Linux, please visit the Red Hat, Inc. website. The documentation for Red Hat Enterprise Linux is available on the Red Hat, Inc. website.
If you are the website administrator:
You may now add content to the webroot directory. Note that until you do so, people visiting your website will see this page, and not your content.
For systems using the Apache HTTP Server: You may now add content to the directory /var/www/html/. Note that until you do so, people visiting your website will see this page, and not your content. To prevent this page from ever being used, follow the instructions in the file /etc/httpd/conf.d/welcome.conf.
For systems using NGINX: You should now put your content in a location of your choice and edit the root configuration directive in the nginx configuration file /etc/nginx/nginx.conf.
Red Hat Enterprise Linux
NGINX
Type here to search
8:22 PM
13-Apr-26